

Technical Report: Forensic Investigation of Unauthorized Privilege Escalation

Project Category: Insider Threat Detection & IAM Governance

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Incident Overview & Executive Summary

At Greenwood Technical College, IT flagged some unusual activity in the student records system. One of the staff members, Karen Doyle, a student services administrator, has legitimate access to student records, financial aid info and enrollment details.

But when we dug into the logs, we saw her account was looking at two specific student profiles way more than normal.

Now, those two students aren't random. One of them, Maya Doyle, is her daughter. The other, Josh Evans, is a friend of the family.

The system shows that Maya's grades for two courses mysteriously changed from C+ to A- without any instructor sign-off. At the same time, Josh, who had been struggling academically, suddenly saw his GPA increase just before getting a merit-based scholarship that he previously didn't qualify for.

A faculty member even dropped an anonymous tip, saying they overheard Maya and Josh joking about "knowing the right people" in administration. So, when we cross-checked system logs, guess whose credentials were used to make those changes? Yep—Karen's. When confronted, she denied it and said maybe someone compromised her account.

Now, this is where you come in.

Investigation Scope: This report analyzes the breach, evaluates the ethical and legal implications, and provides remediation strategies for the following security pillars:

1. Karen technically had access to student records, but should she have been looking at or modifying her daughter's and friend's information? Where's the ethical line here?
2. Let's say Karen thought she was just helping helping her daughter get into a good grad school, helping a struggling student get financial aid. Does that make it okay? Why or why not?
3. If you were designing security policies for this college, how would you prevent this from happening in the future? What kind of oversight should exist for privileged access?
4. Imagine you're Karen. Your child is struggling academically, and you have the power to "fix" it. What's the ethical way to handle that situation?
5. Karen got caught, but what should happen next? Should the punishment be different if she admitted it versus if she kept denying it until the investigation proved she was lying?

1. Karen technically had access to student records, but should she have been looking at or modifying her daughter's and friend's information? Where's the ethical line here?

The Ethical Line:

In cybersecurity, Karen had the rights (the ability to log in) but she did not have authorization for these actions. We learn about the Principle of Least Privilege (PoLP), which says staff should only see data they need for their specific job. Even though

Karen is an admin, looking at her daughter's or friend's files is a major Conflict of Interest. **The ethical line is crossed when personal feelings interfere with professional duties.** In a safe work environment, having access isn't a "free pass" to look at whatever you want; it is a tool for work tasks only. By checking Maya's and Josh's records, she broke the trust the college gave her. In the CIA Triad, this is a failure of "Integrity". Even if she didn't change anything, just "snooping" is a breach of privacy. Professionals should always ask a different coworker to handle files for people they know personally to keep things fair.

2. Let's say Karen thought she was just helping—helping her daughter get into a good grad school, helping a struggling student get financial aid. Does that make it okay? Why or why not?

The "Helping" Justification:

Even if Karen had "good intentions," it does not make her actions right. In security, intent doesn't matter as much as the result. Karen might think she was just being a good mother, but she actually committed "Unauthorized Data Modification" By changing Maya's grades, she made the hard work of other students less valuable. By giving Josh a scholarship he didn't earn, **she basically stole money from a student who actually deserved it.** This is a problem of fairness. In cybersecurity, we look at the result: a compromised system. If every staff member "helped" their friends, the college's degrees would become worthless because no one would trust the grades. Ethical behavior means using the right path—like getting a tutor for Maya—instead of breaking security rules to cheat. Helping two people by lying actually, hurts the reputation of the whole school and every other student.

3. If you were designing security policies for this college, how would you prevent this from happening in the future? What kind of oversight should exist for privileged access?

Policy Design:

To stop this from happening again, I would use several security controls from the Security+ syllabus. First, I would use "**Separation of Duties**" (SoD). This means the person who sees a grade shouldn't be the only one who can change it, a second person must approve it. Technically, the **system should have "Rule-Based Access Control"** to flag an admin if they try to open a file of someone with the same last name or address. We also need "**Continuous Monitoring**" and a **SIEM system**. If an account suddenly looks at one student's profile many times, the system should alert a manager. Finally, the college needs a clear "**Acceptable Use Policy**" (AUP). Every employee must sign this and go through "**Security Awareness Training**" every year. By using these layers, we create "**Defense in Depth**" making it much harder for an insider to abuse their power without getting caught.

4. Imagine you're Karen. Your child is struggling academically, and you have the power to "fix" it. What's the ethical way to handle that situation?

Ethical Conflict:

If I were Karen, the ethical way to help my child is to be a parent, not a hacker. Using work power to fix a grade is a huge risk to her job and her daughter's future. The right thing to do is to be "transparent". Karen should have told her boss immediately when her daughter started at the college. She should have asked the IT team to block her own access to Maya's Student ID so there is no temptation. To help her daughter, she should use official resources: talking to teachers, finding a tutor, or using the college's study center. By cheating, she isn't helping Maya learn; she is teaching her that you don't need to work hard if you know the right people. Real success has to be earned. Ethics is about doing the right thing even when you think no one is watching the logs.

5. Karen got caught, but what should happen next? Should the punishment be different if she admitted it versus if she kept denying it until the investigation proved she was lying?

Accountability:

Karen's choice to lie about the logs shows a total lack of "Accountability". In our security studies, we talk about "non-repudiation". This means that because Karen used her own username and password, she cannot easily claim it wasn't her. The system logs are proof. Her punishment should be much worse because she lied instead of admitting her mistake. If she had told the truth, it might look like a "bad choice" and result in a warning. But because she lied until the evidence proved she was wrong, she showed that she has no integrity. This makes her a permanent "Insider Threat" who can never be trusted with data again. In the professional world, **changing financial or school records for personal gain is fraud**. She should be fired and possibly face legal trouble. This sends a "Deterrent" message to other employees that the rules must be followed.